

전국기능경기대회 채점기준

1. 채점 시 유의사항

직 종 명

클라우드컴퓨팅

※ 다음 사항을 유의하여 채점하시오.

- 1) AWS의 지역은 ap-northeast-2를 사용합니다.
- 2) 웹페이지 접근은 크롬이나 파이어폭스를 이용합니다.
- 3) 웹페이지에서 언어에 따라 문구가 다르게 보일 수 있습니다.
- 4) shell에서의 명령어의 출력은 버전에 따라 조금 다를 수 있습니다.
- 5) 채점은 문항 순서대로 진행해야 합니다.
- 6) 부분 점수가 따로 없는 문항은 모두 맞아야 점수로 인정됩니다.
- 7) 리소스의 정보를 읽어오는 채점항목은 기본적으로 스크립트 결과를 통해 채점을 진행하며, 만약 선수가 이의가 있다면 명령어를 직접 입력하여 확인해볼 수 있습니다.
- 8) (예상 출력)은 바로 이전 (명령어 입력)의 예상 출력을 의미합니다.
- 9) 채점 시에는 별도로 제공한 채점 스크립트(mark.sh)를 실행하여 채점할 수 있습니다. 다만, 선수가 직접 입력을 원할 경우 채점기준표에 명시된 명령어 그대로 입력하여 채점할 수 있습니다.
- 10) 제출 전 mark-sg의 경우 선수 임의로 구성해두도록 합니다.
- 11) 모든 채점 사항은 Cloudshell에 wsc2026-skills-app-sub-a 서버넷과 사전구성한 mark-sg를 연결한 후 진행합니다.
- 12) 11-1 채점을 실행하여 생긴 파드들은 채점이 끝난 후 삭제하도록 합니다.
- 13) 아래 명령어를 사전입력합니다.

```
aws configure set default.region ap-northeast-2
```

```
ACCOUNT_ID=$(aws sts get-caller-identity --query "Account" --output text)
```

```
VPC_ID=$(aws ec2 describe-vpcs --filters "Name=tag:Name,Values=wsc2026-skills-vpc" --query "Vpcs[0].VpcId" --output text)
```

```
BUCKET=$(aws s3api list-buckets --query "Buckets[?contains(Name,'wsc2026-static')].Name" --output text)
```

```
aws eks update-kubeconfig --name wsc2026-eks-cluster --region ap-northeast-2 2>/dev/null
```

```
check_kms() {
```

```
    local ALIAS=$1 ACTUAL_ARN=$2; local EXPECTED_ARN=$(aws kms describe-key --key-id "alias/${ALIAS}" --query "KeyMetadata.Arn" --output text 2>/dev/null); local KEY_ID=$(aws kms describe-key --key-id "alias/${ALIAS}" --query "KeyMetadata.KeyId" --output text 2>/dev/null)
    if [ -z "$KEY_ID" ] || [ "$KEY_ID" = "None" ]; then echo "KMS ${ALIAS}: FAIL (key not found)";
    return; fi
```

```
    if [ "$EXPECTED_ARN" != "$ACTUAL_ARN" ]; then echo "KMS ${ALIAS}: FAIL (wrong key)"; return; fi
```

```
    POLICY=$(aws kms get-key-policy --key-id "$KEY_ID" --policy-name default --output text 2>/dev/null)
```

```
    if echo "$POLICY" | grep -q '"kms:W*"' ; then echo "KMS ${ALIAS}: FAIL (kms:*)"; elif echo "$POLICY" | grep -q ':root' ; then echo "KMS ${ALIAS}: FAIL (root)"; else echo "KMS ${ALIAS}: PASS"; fi
```

```
}
```

※ 채점기준 양식은 반드시, 양식에 맞추어 작성해야 합니다.(채점사이트 입력에 필요)

2. 채점기준표

1) 주요항목별 배점			직 종 명		클라우드컴퓨팅			
과제 번호	일련 번호	주요항목	배점	채점방법		채점시기		비고
				독립	합의	경기 진행중	경기 종료후	
제1과제	1	Networking	2.0	○			○	
	2	Database	1.3	○			○	
	3	Container Registry	1.2	○			○	
	4	Container Orchestration	3.0	○			○	
	5	Deployment	6.5	○			○	
	6	S3	1.0	○			○	
	7	Lambda	3.0	○			○	
	8	ALB	1.3	○			○	
	9	CloudFront	4.0	○			○	
	10	WAF	1.5	○			○	
	11	Observability	5.2	○			○	
합 계			30					

2) 채점방법 및 기준

과제 번호	일 련 번 호	주요항목	일련 번호	세부항목(채점방법)	배점
제1과제	1	Networking	1	VPC	1.0
			2	Routing	1.0
	2	Database	1	DynamoDB Table	1.3
	3	Container Registry	1	ECR Container	1.2
	4	Container Orchestration	1	EKS Cluster	1.5
			2	NodeGroup	1.0
			3	EKS IAM	0.5
	5	Deployment	1	Resource Status	1.5
			2	Scheduling & Resources	1.0
			3	Probes & ConfigMap	1.0
			4	Service & Pod Placement	1.5
			5	Pod Identity	1.5
	6	S3	1	S3 Bucket	1.0
	7	Lambda	1	Lambda Functions	1.5
			2	Lambda IAM	1.5
	8	Load Balancer	1	Load Balancer	1.3
	9	CloudFront	1	CloudFront Distribution	1.2
			2	Cache Policy	1.3
			3	POST & GET E2E	1.5
	10	WAF	1	WAF Protection	1.5
	11	Observability	1	Observability Deploy	1.2
			2	Grafana Datasource	1.0
			3	Grafana Dashboard	1.5
			4	Alert Firing	1.5
합계					30

순번	채점 항목															
1-1	1) <code>aws ec2 describe-vpcs --vpc-ids "\${VPC_ID}" --query "Vpcs[0].CidrBlock" --output text;</code> <code>aws ec2 describe-subnets --filters "Name=vpc-id,Values=\${VPC_ID}" --query "Subnets[0].{Name:Tags[?Key=='Name'].Value[0],Cidr:CidrBlock}" --output text</code> 2) VPC와 서브넷의 이름이 정확하게 일치하며 VPC와 서브넷에 지정한 CIDR가 부여되어있는지 확인합니다. 192.168.0.0/16 192.168.1.0/24 wsc2026-skills-hub-sub-a 192.168.10.0/24 wsc2026-skills-hub-sub-b 192.168.2.0/24 wsc2026-skills-app-sub-a 192.168.20.0/24 wsc2026-skills-app-sub-b															
1-2	1) <code>IGW_ID=\$(aws ec2 describe-internet-gateways --filters "Name=attachment.vpc-id,Values=\${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-igw" --query "InternetGateways[0].InternetGatewayId" --output text); echo "IGW: \$IGW_ID"</code> <code>NAT_A_ID=\$(aws ec2 describe-nat-gateways --filter "Name=vpc-id,Values=\${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-nat-a" "Name=state,Values=available" --query "NatGateways[0].NatGatewayId" --output text);</code> <code>NAT_B_ID=\$(aws ec2 describe-nat-gateways --filter "Name=vpc-id,Values=\${VPC_ID}" "Name=tag:Name,Values=wsc2026-skills-nat-b" "Name=state,Values=available" --query "NatGateways[0].NatGatewayId" --output text); echo "NAT-A: \$NAT_A_ID"; echo "NAT-B: \$NAT_B_ID"</code> <code>aws ec2 describe-route-tables --filters "Name=vpc-id,Values=\${VPC_ID}" "Name=tag:Name,Values=*" --query "RouteTables[0].{Name:Tags[?Key=='Name'].Value[0],Route:Routes[?DestinationCidrBlock=='0.0.0.0/0']}[0]" --output text</code> 2) 실행 결과가 강조된 텍스트와 동일하며, hub-rtb에는 igw의 ID가, app-rtb-a에는 NAT-A의 ID가, app-rtb-b에는 NAT-B의 ID가 서로 일치하게 연결되어 있는지 확인합니다. IGW: igw-0a85cd67732857621 NAT-A: nat-071e3d4820939525d NAT-B: nat-03f60025e555f65bb wsc2026-skills-app-rtb-b <table><tr><td>ROUTE</td><td>0.0.0.0/0</td><td>nat-03f60025e555f65bb</td><td>CreateRoute</td><td>active</td></tr></table> wsc2026-skills-hub-rtb <table><tr><td>ROUTE</td><td>0.0.0.0/0</td><td>igw-0a85cd67732857621</td><td>CreateRoute</td><td>active</td></tr></table> wsc2026-skills-app-rtb-a <table><tr><td>ROUTE</td><td>0.0.0.0/0</td><td>nat-071e3d4820939525d</td><td>CreateRoute</td><td>active</td></tr></table>	ROUTE	0.0.0.0/0	nat-03f60025e555f65bb	CreateRoute	active	ROUTE	0.0.0.0/0	igw-0a85cd67732857621	CreateRoute	active	ROUTE	0.0.0.0/0	nat-071e3d4820939525d	CreateRoute	active
ROUTE	0.0.0.0/0	nat-03f60025e555f65bb	CreateRoute	active												
ROUTE	0.0.0.0/0	igw-0a85cd67732857621	CreateRoute	active												
ROUTE	0.0.0.0/0	nat-071e3d4820939525d	CreateRoute	active												

순번	채점 항목
2-1	1) aws dynamodb describe-table --table-name wsc2026-book-table --query "Table.[KeySchema[0].AttributeName,BillingModeSummary.BillingMode,SSEDescription.SSEType,DeletionProtectionEnabled,GlobalSecondaryIndexes[0].KeySchema[0].AttributeName]" --output text 2>/dev/null aws dynamodb describe-continuous-backups --table-name wsc2026-book-table --query "ContinuousBackupsDescription.[PointInTimeRecoveryDescription.PointInTimeRecoveryStatus,PointInTimeRecoveryDescription.RecoveryPeriodInDays,ContinuousBackupsStatus]" --output text 2>/dev/null aws dynamodb get-resource-policy --resource-arn "arn:aws:dynamodb:ap-northeast-2:\${ACCOUNT_ID}:table/wsc2026-book-table" --query "Policy" --output text 2>/dev/null python3 -c "import sys,json:[print(f'{s.get(W\"ActionW\",W\"W\")} : {s.get(W\"PrincipalW\",{ }).get(W\"AWSW\",W\"W\").split(W\"/W\")[-1]})' for s in json.loads(sys.stdin.read()).get('Statement',[])]" 2>/dev/null echo "No resource policy" check_kms "wsc2026-db-kms" "\$(aws dynamodb describe-table --table-name wsc2026-book-table --query 'Table.SSEDescription.KMSMasterKeyArn' --output text 2>/dev/null)" 2) 실행 결과가 아래와 정확히 일치해야 합니다. 단, 강조된 부분의 출력 순서는 무관합니다. <pre>client_id PAY_PER_REQUEST KMS True booking_id ENABLED 35 ENABLED</pre> <u>dynamodb:PutItem : wsc2026-book-pod-role</u> <u>dynamodb:Query : wsc2026-book-function-role</u> KMS wsc2026-db-kms: PASS
3-1	1) aws ecr describe-repositories --repository-names wsc2026-book-ecr --query "repositories[0].[imageScanningConfiguration.scanOnPush,imageTagMutability,imageTagMutabilityExclusionFilters[0].filter,encryptionConfiguration.encryptionType]" --output text 2>/dev/null; aws ecr list-images --repository-name wsc2026-book-ecr --query "imageIds[].imageTag" --output text 2>/dev/null check_kms "wsc2026-ecr-kms" "\$(aws ecr describe-repositories --repository-names wsc2026-book-ecr --query 'repositories[0].encryptionConfiguration.kmsKey' --output text 2>/dev/null)" 2) 실행 결과가 아래와 동일한지 확인합니다. <pre>True MUTABLE_WITH_EXCLUSION v1* KMS v1.0.0</pre> KMS wsc2026-ecr-kms: PASS

순번	채점 항목
4-1	1) <code>aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.[version,status,resourcesVpcConfig.endpointPublicAccess,resourcesVpcConfig.endpointPrivateAccess,logging.clusterLogging[0].enabled]" --output text 2>/dev/null</code> <code>CLUSTER_SG=\$(aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.resourceVpcConfig.clusterSecurityGroupId" --output text 2>/dev/null); ANYIP=\$(aws ec2 describe-security-groups --group-ids "\$CLUSTER_SG" --query "SecurityGroups[0].IpPermissions[?IpProtocol=='-1'].IpRanges[].CidrIp" --output text 2>/dev/null); if [-n "\$ANYIP"]; then echo "SG: FAIL"; else echo "SG: PASS"; fi</code> <code>check_kms "wsc2026-eks-kms" "\$(aws eks describe-cluster --name wsc2026-eks-cluster --query 'cluster.encryptedConfig[0].provider.keyArn' --output text 2>/dev/null)"</code> 2) 실행 결과가 아래와 동일한지 확인합니다. <pre> 1.35 ACTIVE False True True SG: PASS wsc2026.skills.local KMS wsc2026-eks-kms: PASS </pre>
4-2	1) <code>NG1=\$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-addon-nodegroup --query "nodegroup.[nodegroupName,instanceTypes[0]]" --output text 2>/dev/null); echo "\$NG1 wsc2026/node=addon"</code> <code>ADDON_COUNT=\$(kubectl get nodes -l wsc2026/node=addon --no-headers --request-timeout=10s 2>/dev/null wc -l)</code> <code>if ["\$ADDON_COUNT" -ge 1]; then echo "Addon Nodes: PASS (\$ADDON_COUNT)"; else echo "Addon Nodes: FAIL"; fi</code> <code>NG2=\$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-workload-ng --query "nodegroup.[nodegroupName,instanceTypes[0]]" --output text 2>/dev/null); echo "\$NG2 wsc2026/node=application"</code> <code>WORK_COUNT=\$(kubectl get nodes -l wsc2026/node=application --no-headers --request-timeout=10s 2>/dev/null wc -l)</code> <code>if ["\$WORK_COUNT" -ge 1]; then echo "Workload Nodes: PASS (\$WORK_COUNT)"; else echo "Workload Nodes: FAIL"; fi</code> 2) 실행 결과가 아래와 동일한지 확인합니다. <pre> wsc2026-addon-nodegroup t3.medium wsc2026/node=addon Addon Nodes: PASS (2) wsc2026-workload-ng t3.medium wsc2026/node=application Workload Nodes: PASS (2) </pre>

순번	채점 항목
4-3	1) CR=\$(aws eks describe-cluster --name wsc2026-eks-cluster --query "cluster.roleArn" --output text 2>/dev/null awk -F/ '{print \$NF}'); AR=\$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-addon-nodegroup --query "nodegroup.nodeRole" --output text 2>/dev/null awk -F/ '{print \$NF}'); WR=\$(aws eks describe-nodegroup --cluster-name wsc2026-eks-cluster --nodegroup-name wsc2026-workload-ng --query "nodegroup.nodeRole" --output text 2>/dev/null awk -F/ '{print \$NF}') for PAIR in "Cluster Role:\$CR" "Addon Node Role:\$AR" "Workload Node Role:\$WR"; do LABEL="\${PAIR%:*}"; ROLE="\${PAIR#*:}"; ADMIN=\$(aws iam list-attached-role-policies --role-name "\$ROLE" --query "AttachedPolicies[?PolicyName=='AdministratorAccess'].PolicyName" --output text 2>/dev/null); if [-n "\$ADMIN"]; then echo "\$LABEL: FAIL"; else echo "\$LABEL: PASS"; fi; done 2) 실행 결과가 아래와 동일한지 확인합니다. Cluster Role: PASS Addon Node Role: PASS Workload Node Role: PASS
5-1	1) kubectl get deploy,svc,ingress,pdb -n wsc2026 --no-headers 2>/dev/null awk ' /^deployment/ { split(\$2,a,"/"); print "Deployment: " (a[1]==a[2] && a[1]=="2" ? "PASS" : "FAIL") } /^service/ { print "Service: PASS" } /^ingress/ { print "Ingress: " (\$4 ~ /W.elbW.amazonawsW.com/ ? "PASS (" \$4)" : "FAIL") } /^poddisrupt/ { print "PDB: " (\$2=="1" ? "PASS" : "FAIL") } , 2) 실행결과가 아래와 동일한지 확인하며, Ingress의 ALB DNS는 wsc2026-app-alb가 포함되었는지 확인합니다. Deployment: PASS Service: PASS Ingress: PASS (wsc2026-app-alb-829903010.ap-northeast-2.elb.amazonaws.com) PDB: PASS
5-2	1) kubectl get deploy wsc2026-book-deploy -n wsc2026 -o jsonpath='{.spec.replicas:{.spec.replicas} node:{.spec.template.spec.nodeSelector.wsc2026/node} topo:{.spec.template.spec.topologySpreadConstraints[0].topologyKey} cpu:{.spec.template.spec.containers[0].resources.requests.cpu} mem:{.spec.template.spec.containers[0].resources.requests.memory} ' 2>/dev/null 2) 실행 결과가 아래와 동일한지 확인합니다. replicas:2 node:application topo:topology.kubernetes.io/zone cpu:250m mem:512Mi

순번	채점 항목
5-3	1) <code>kubectl get deploy wsc2026-book-deploy -n wsc2026 -o jsonpath='{startup:{.spec.template.spec.containers[0].startupProbe.httpGet.path}:{.spec.template.spec.containers[0].startupProbe.httpGet.port} readiness:{.spec.template.spec.containers[0].readinessProbe.httpGet.path}:{.spec.template.spec.containers[0].readinessProbe.httpGet.port} liveness:{.spec.template.spec.containers[0].livenessProbe.httpGet.path}:{.spec.template.spec.containers[0].livenessProbe.httpGet.port}}' 2>/dev/null; echo; kubectl get configmap book-config -n wsc2026 -o jsonpath='{.data}' 2>/dev/null; echo</code> 2) 실행 결과가 아래와 동일한지 확인합니다. startup:/health:8080 readiness:/health:8080 liveness:/health:8080 {"AWS_REGION":"ap-northeast-2","TABLE_NAME":"wsc2026-book-table"}
5-4	1) <code>WORKLOAD_NODES=\$(kubectl get nodes -l wsc2026/node=application --no-headers -o custom-columns='NAME:.metadata.name' 2>/dev/null)</code> <code>kubectl get pods -n wsc2026 -o custom-columns='NAME:.metadata.name,STATUS:.status.phase,NODE:.spec.nodeName' --no-headers 2>/dev/null while read NAME STATUS NODE; do</code> <code>echo "\$NAME: \$(echo "\$WORKLOAD_NODES" grep -q "^\$NODE\$" && echo PASS echo FAIL)"</code> <code>done</code> 2) 실행 결과에서 wsc2026-book-deploy가 포함된 텍스트가 2개 출력되며, 모두 PASS인지 확인합니다. wsc2026-book-deploy-7b88597d86-d6l72: PASS wsc2026-book-deploy-7b88597d86-z8nkt: PASS
5-5	1) <code>PI_SA=\$(aws eks list-pod-identity-associations --cluster-name wsc2026-eks-cluster --namespace wsc2026 --query 'associations[0].serviceAccount' --output text 2>/dev/null)</code> <code>PI_ROLE=\$(aws eks list-pod-identity-associations --cluster-name wsc2026-eks-cluster --namespace wsc2026 --query 'associations[0].associationId' --output text 2>/dev/null)</code> <code>["\$PI_SA" = "wsc2026-book-sa"] && echo "Pod Identity SA: PASS (\$PI_SA)" echo "Pod Identity SA: FAIL (\$PI_SA)"</code> <code>ROLE_NAME=\$(aws eks describe-pod-identity-association --cluster-name wsc2026-eks-cluster --association-id "\$PI_ROLE" --query 'association.roleArn' --output text 2>/dev/null awk -F/ '{print \$NF}')</code> <code>MANAGED_ACTIONS=\$(aws iam list-attached-role-policies --role-name "\$ROLE_NAME" --query 'AttachedPolicies[].PolicyArn' --output text 2>/dev/null tr 'Wt' 'Wn' while read arn;do</code> <code>VERSION=\$(aws iam get-policy --policy-arn "\$arn" --query 'Policy.DefaultVersionId' --output text 2>/dev/null);aws iam get-policy-version --policy-arn "\$arn" --version-id "\$VERSION" --query 'PolicyVersion.Document.Statement[].Action' --output text</code> <code>2>/dev/null;done)</code> <code>echo "\$MANAGED_ACTIONS" grep -q dynamodb:PutItem && ! echo "\$MANAGED_ACTIONS" grep -qE '^W*\$' && echo "Pod Identity Role: PASS" echo "Pod Identity Role: FAIL"</code> 2) 실행 결과가 아래와 동일한지 확인합니다. Pod Identity SA: PASS (wsc2026-book-sa) Pod Identity Role: PASS

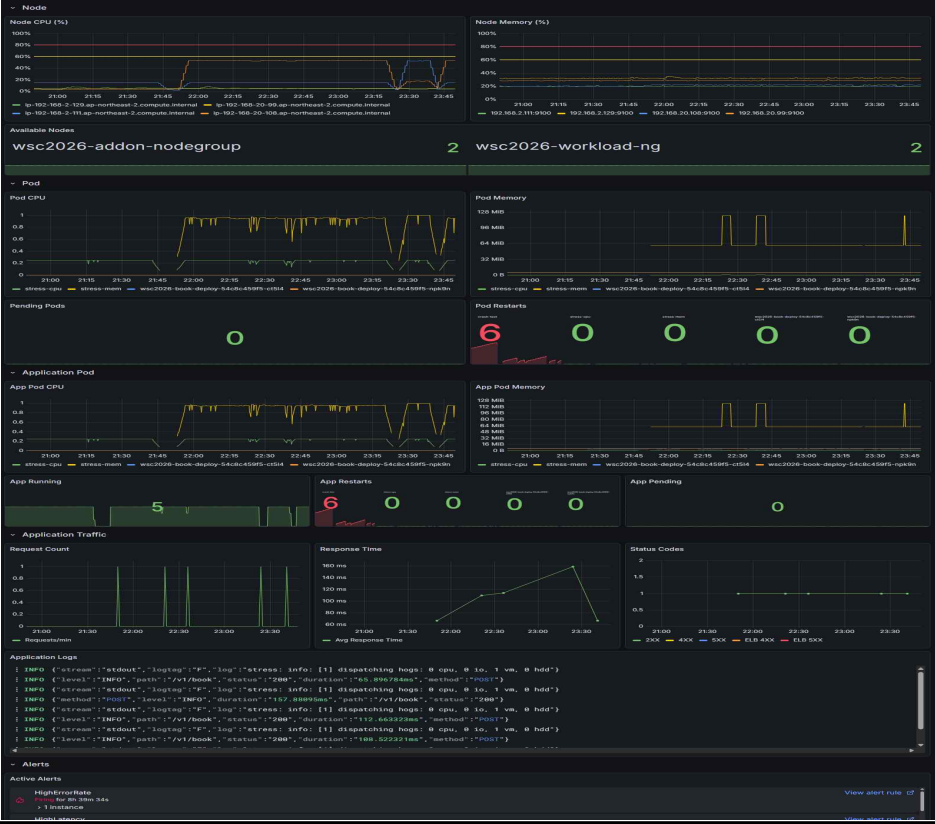
순번	채점 항목
6-1	<pre> 1) echo "\$BUCKET"; aws s3api get-public-access-block --bucket "\$BUCKET" --query "PublicAccessBlockConfiguration.[BlockPublicAcls,BlockPublicPolicy,IgnorePublicAcls,RestrictPublicBuckets]" --output text 2>/dev/null; aws s3api get-bucket-encryption --bucket "\$BUCKET" --query "ServerSideEncryptionConfiguration.Rules[0].{SSE:ApplyServerSideEncryptionByDefault.SSEAlgorithm,BucketKey:BucketKeyEnabled}" --output text 2>/dev/null; aws s3api list-objects --bucket "\$BUCKET" --prefix "static/" --query "Contents[?Size>W`OW`].Key" --output text 2>/dev/null check_kms "wsc2026-bucket-kms" "\$(aws s3api get-bucket-encryption --bucket "\$BUCKET" --query 'ServerSideEncryptionConfiguration.Rules[0].ApplyServerSideEncryptionByDefault.KMSMasterKeyID' --output text 2>/dev/null)" BUCKET_KMS_ARN=\$(aws kms describe-key --key-id "alias/wsc2026-bucket-kms" --query "KeyMetadata.Arn" --output text 2>/dev/null) echo "S3 Object KMS Check:"; for OBJ in \$(aws s3api list-objects --bucket "\$BUCKET" --prefix "static/" --query "Contents[].Key" --output text 2>/dev/null); do KEY_ID=\$(aws s3api head-object --bucket "\$BUCKET" --key "\$OBJ" --query "SSEKMSKeyId" --output text 2>/dev/null) if ["\$KEY_ID" = "\$BUCKET_KMS_ARN"]; then echo " \$OBJ: PASS"; else echo " \$OBJ: FAIL (\$KEY_ID)"; fi done 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-static-<임의 영문 4자리>-<선수 비번호>-bucket True True True True True aws:kms static/index.html static/main.jpeg KMS wsc2026-bucket-kms: PASS S3 Object KMS Check: —static/: PASS static/index.html: PASS static/main.jpeg: PASS </pre>
7-1	<pre> 1) aws lambda get-function --function-name wsc2026-book-get-function --query "Configuration.{Runtime:Runtime,Env:Environment.Variables.TABLE_NAME}" --output text 2>/dev/null check_kms "wsc2026-function-kms" "\$(aws lambda get-function --function-name wsc2026-book-get-function --query 'Configuration.KMSKeyArn' --output text 2>/dev/null)" 2) 실행 결과가 아래와 동일한지 확인하며, TABLE_NAME은 평문이 아닌 KMS로 암호화된 텍스트가 출력되는지 확인합니다. { "Name": "wsc2026-book-get-function", "Runtime": "python3.12" } { "TABLE_NAME": "AQICAHjHYzx8HkYlOX8fnCJbPqPanGMX7gxAX57CFiSG1wT1wEpFZotnxm3...." } KMS wsc2026-function-kms: PASS </pre>

순번	채점 항목
7-2	1) <code>LAMBDA_ROLE=\$(aws lambda get-function --function-name wsc2026-book-get-function --query "Configuration.Role" --output text 2>/dev/null awk -F/ '{print \$NF}')</code>; echo "\$LAMBDA_ROLE"; <code>POLICIES=\$(aws iam list-attached-role-policies --role-name "\$LAMBDA_ROLE" --query "AttachedPolicies[].PolicyName" --output text 2>/dev/null)</code>; echo "\$POLICIES"; if echo "\$POLICIES" grep -q "AdministratorAccess"; then echo "Role: FAIL (Admin)"; else echo "Role: PASS"; fi <code>POLICY_ARN=\$(aws iam list-attached-role-policies --role-name "\$LAMBDA_ROLE" --query "AttachedPolicies[?PolicyName!='AWSLambdaBasicExecutionRole'].PolicyArn [0]" --output text 2>/dev/null)</code>; <code>ACTIONS=\$(aws iam get-policy-version --policy-arn "\$POLICY_ARN" --version-id \$(aws iam get-policy --policy-arn "\$POLICY_ARN" --query "Policy.DefaultVersionId" --output text 2>/dev/null) --query "PolicyVersion.Document.Statement[].Action" --output text 2>/dev/null)</code> if echo "\$ACTIONS" grep -q "dynamodb:Query" && ! echo "\$ACTIONS" grep -q 'W*'; then echo "Policy: PASS"; else echo "Policy: FAIL"; fi 2) 실행 결과가 아래와 동일한지 확인합니다. wsc2026-book-function-role wsc2026-book-function-policy Role: PASS Policy: PASS
8-1	1) <code>aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].Scheme" --output text 2>/dev/null</code>; <code>ALB_SGS=\$(aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].SecurityGroups[]" --output text 2>/dev/null)</code>; <code>aws ec2 describe-security-groups --group-ids \$ALB_SGS --query "SecurityGroups[].GroupName" --output text 2>/dev/null</code> <code>ALB_DNS=\$(aws elbv2 describe-load-balancers --names wsc2026-app-alb --query "LoadBalancers[0].DNSName" --output text 2>/dev/null)</code>; <code>ALB_CODE=\$(curl -s -o /dev/null -w "%{http_code}" --max-time 5 "http://\$ALB_DNS/" 2>/dev/null)</code>; if ["\$ALB_CODE" = "000"]; then echo "ALB direct: BLOCKED"; else echo "ALB direct: FAIL (\$ALB_CODE)"; fi 2) 실행 결과가 아래와 동일한지 확인합니다. internet-facing wsc2026-app-alb-sg ALB direct: BLOCKED
9-1	1) <code>CF_ARN=\$(aws resourcegroupstaggingapi get-resources --tag-filters Key=Name,Values=wsc2026-cdn --resource-type-filters cloudfront --region us-east-1 --query "ResourceTagMappingList[0].ResourceARN" --output text 2>/dev/null)</code>; <code>CF_ID=\$(echo "\$CF_ARN" awk -F/ '{print \$NF}')</code>; <code>CF_DOMAIN=\$(aws cloudfront get-distribution --id "\$CF_ID" --region us-east-1 --query "Distribution.DomainName" --output text 2>/dev/null)</code>; <code>curl -s -o /dev/null -w "%{http_code}" "https://\$CF_DOMAIN/" 2>/dev/null</code>; echo 2) 실행 결과에서 Cloudfront 도메인과 200이 출력되는지 확인합니다. d1ain3jwa2akkh.cloudfront.net : 200

순번	채점 항목
9-2	1) DISABLED="4135ea2d-6df8-44a3-9df3-4b5a84be39ad"; OPTIMIZED="658327ea-f89d-4fab-a63d-7e88639e58f6"; DEFAULT_CP=\$(aws cloudfront get-distribution --id "\$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.DefaultCacheBehavior.CachePolicyId" --output text 2>/dev/null); if ["\$DEFAULT_CP" = "\$OPTIMIZED"]; then echo "S3: CachingOptimized"; else echo "S3: FAIL"; fi for B in \$(aws cloudfront get-distribution --id "\$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.CacheBehaviors.Items[].PathPattern" --output text 2>/dev/null); do CP=\$(aws cloudfront get-distribution --id "\$CF_ID" --region us-east-1 --query "Distribution.DistributionConfig.CacheBehaviors.Items[?PathPattern=='\${B}'].CachePolicyId[0]" --output text 2>/dev/null); if ["\$CP" = "\$DISABLED"]; then echo "ALB/Lambda: CachingDisabled"; break; fi; done 2) 실행 결과가 아래와 동일한지 확인합니다. S3: CachingOptimized ALB/Lambda: CachingDisabled
9-3	1) BOOKING_ID=\$(curl -s -X POST "https://\${CF_DOMAIN}/booking" -H "Content-Type: application/json" -d '{"client_id":"MARK001","username":"Marker","email":"mark@test.com","concert_name":"TestConcert"}' 2>/dev/null python3 -c "import sys,json;print(json.load(sys.stdin).get('booking_id',''))" 2>/dev/null); echo "POST booking_id: \$BOOKING_ID" curl -s "https://\${CF_DOMAIN}/v1/book?booking_id=\${BOOKING_ID}" 2>/dev/null; echo 2) 필드 순서(client_id, username, email, concert_name, created_at)가 차례대로 일치하는지, 그리고 created_at 시간이 스크립트 실행 시간 기준 1분 이내인지 확인합니다. POST booking_id: NWOEMIU2 {"client_id": "MARK001", "username": "Marker", "email": "mark@test.com", "concert_name": "TestConcert", "created_at": "2026-05-15 00:13:05 KST"}
10-1	1) WAF_NAME=\$(aws wafv2 list-web-acls --scope CLOUDFRONT --region us-east-1 --query "WebACLs[?contains(Name, 'wsc2026')].Name" --output text 2>/dev/null) echo "WAF Name: \$WAF_NAME" SQLI=\$(curl -s -o /dev/null -w '%{http_code}' --max-time 5 "https://\${CF_DOMAIN}/v1/book?booking_id=1'%200R%201=t--"); echo "SQLi: \$SQLI" XSS=\$(curl -s -o /dev/null -w '%{http_code}' --max-time 5 "https://\${CF_DOMAIN}/v1/book?booking_id=<script>alert(1)</script>"); echo "XSS: \$XSS" (for i in \$(seq 1 15); do for j in \$(seq 1 25); do curl -s -o /dev/null --max-time 3 "https://\${CF_DOMAIN}/" & done; sleep 0.4; done; wait) >/dev/null 2>&1; WAF_ID=\$(aws wafv2 list-web-acls --scope CLOUDFRONT --region us-east-1 --query "WebACLs[?contains(Name, 'wsc2026')].Id" --output text 2>/dev/null); RATE_LIMIT=\$(aws wafv2 get-web-acl --scope CLOUDFRONT --region us-east-1 --name "\$WAF_NAME" --id "\$WAF_ID" --query "WebACL.Rules[?Statement.RateBasedStatement].Statement.RateBasedStatement.Limit" --output text 2>/dev/null); if [-n "\$RATE_LIMIT"] && ["\$RATE_LIMIT" -le 200] 2>/dev/null; then echo "Rate: PASS (403)"; else echo "Rate: FAIL (-)"; fi 2) 실행 결과가 아래와 동일한지 확인합니다. WAF Name: wsc2026-waf SQLi: 403 XSS: 403 Rate: PASS (403)

순번	채점 항목
11-1	<pre> 1) SVC_IP=\$(kubectl get svc -n wsc2026 -o jsonpath='{.items[0].spec.clusterIP}' 2>/dev/null); kubectl + run not-ready --image=busybox --restart=Always -n wsc2026 --overrides='{ "spec": { "tolerations": [{ "o perator": "Exists" }], "nodeSelector": { "wsc2026/node": "application" }, "containers": [{ "name": "not-ready", "image": "busybox", "readinessProbe": { "httpGet": { "path": "/health", "port": 80 }, "periodSeconds": 3 }, "comma nd": ["sh", "-c", "sleep 3600"]]] } }' &>/dev/null; kubectl run error-gen --image=curlimages/curl --resta rt=Never -n wsc2026 --overrides='{ "spec": { "tolerations": [{ "operator": "Exists" }], "nodeSelector": { "wsc 2026/node": "application" } } }' -- sh -c "while true; do curl -s -o /dev/null http://'\$SVC_IP'/nonexi st; sleep 0.1; done" &>/dev/null; kubectl run latency-gen --image=curlimages/curl --restart=Never -n wsc2026 --overrides='{ "spec": { "tolerations": [{ "operator": "Exists" }], "nodeSelector": { "wsc2026/node": " application" } } }' -- sh -c "while true; do curl -s -o /dev/null http://'\$SVC_IP'/delay?ms=5000; sle ep 0.2; done" &>/dev/null kubectl run crash-test --image=busybox --restart=Always -n wsc2026 --overrides='{ "spec": { "toleration s": [{ "operator": "Exists" }], "nodeSelector": { "wsc2026/node": "application" } } }' -- sh -c 'exit 1' &>/dev /null; kubectl run stress-cpu --image=busybox --restart=Never -n wsc2026 --overrides='{ "spec": { "tole rations": [{ "operator": "Exists" }], "nodeSelector": { "wsc2026/node": "application" }, "containers": [{ "name ": "stress-cpu", "image": "busybox", "resources": { "requests": { "cpu": "250m" }, "limits": { "cpu": "250m" } }, "co mmand": ["sh", "-c", "while true; do :: done"]]] } }' &>/dev/null; kubectl run stress-mem --image=polinux /stress --restart=Never -n wsc2026 --overrides='{ "spec": { "tolerations": [{ "operator": "Exists" }], "node Selector": { "wsc2026/node": "application" }, "containers": [{ "name": "stress-mem", "image": "polinux/stress ", "resources": { "requests": { "memory": "64Mi" }, "limits": { "memory": "64Mi" } }, "command": ["stress", "- vm", "-vm-bytes", "60M", "-vm-keep", "-t", "3600"]]] } }' &>/dev/null sleep 180 GRAFANA_LB=\$(kubectl get svc -n observability -o jsonpath='{range .items[?(@.spec.type=="LoadBalancer r")]}{.status.loadBalancer.ingress[0].hostname}{end}' 2>/dev/null) for p in fluent-bit prometheus grafana; do kubectl get pods -n observability --no-headers --request- timeout=10s 2>/dev/null grep -c "\$p.*Running" xargs -l{} echo "\$p: {}"; done 2) 실행 결과가 1 이상인지 확인합니다. fluent-bit: 4 prometheus: 7 grafana: 1 </pre>
11-2	<pre> 1) echo "Datasources:"; curl -s -u admin:'Skills\$\$@!' "http://\${GRAFANA_LB}/api/datasources" 2>/dev/null python3 -c "import sys,json;[print(f' {d[W"nameW"]} ({d[W"typeW"]})') for d in json.load(sys.stdin)]" 2>/dev/null echo " Grafana unreachable" echo "Dashboards:"; curl -s -u admin:'Skills\$\$@!' "http://\${GRAFANA_LB}/api/search?query=wsc2026" 2>/dev/null python3 -c "import sys,json;[print(f' {d[W"titleW"]}') for d in js on.load(sys.stdin)]" 2>/dev/null echo " Not found" 2) 실행 결과가 아래와 동일한지 확인합니다. Datasources: alertmanager (alertmanager) cloudwatch (cloudwatch) prometheus (prometheus) Dashboards: wsc2026-grafana-dashboard </pre>

순번	채점 항목
11-3	1) 아래 명령을 수동으로 입력합니다. <pre> SVC_IP=\$(kubectl get svc -n wsc2026 -o jsonpath='{.items[0].spec.clusterIP}' 2>/dev/null); kubectl r un not-ready --image=busybox --restart=Always -n wsc2026 --overrides='{ "spec": { "tolerations": [{ "oper ator": "Exists" }], "nodeSelector": { "wsc2026/node": "application" }, "containers": [{ "name": "not-ready", "im age": "busybox", "readinessProbe": { "httpGet": { "path": "/health", "port": 80 }, "periodSeconds": 3 }, "command ": ["sh", "-c", "sleep 3600"]] } } }' &>/dev/null; kubectl run error-gen --image=curlimages/curl --restart =Never -n wsc2026 --overrides='{ "spec": { "tolerations": [{ "operator": "Exists" }], "nodeSelector": { "wsc20 26/node": "application" } } }' -- sh -c "while true; do curl -s -o /dev/null http://'\$SVC_IP'/nonexis t; sleep 0.1; done" &>/dev/null; kubectl run latency-gen --image=curlimages/curl --restart=Never -n wsc2026 --overrides='{ "spec": { "tolerations": [{ "operator": "Exists" }], "nodeSelector": { "wsc2026/node": " application" } } }' -- sh -c "while true; do curl -s -o /dev/null http://'\$SVC_IP'/delay?ms=5000; sle ep 0.2; done" &>/dev/null kubectl run crash-test --image=busybox --restart=Always -n wsc2026 --overrides='{ "spec": { "toleration s": [{ "operator": "Exists" }], "nodeSelector": { "wsc2026/node": "application" } } }' -- sh -c 'exit 1' &>/dev /null; kubectl run stress-cpu --image=busybox --restart=Never -n wsc2026 --overrides='{ "spec": { "tole rations": [{ "operator": "Exists" }], "nodeSelector": { "wsc2026/node": "application" }, "containers": [{ "name ": "stress-cpu", "image": "busybox", "resources": { "requests": { "cpu": "250m" }, "limits": { "cpu": "250m" } }, "co mmand": ["sh", "-c", "while true; do :: done"]] } } }' &>/dev/null; kubectl run stress-mem --image=polinux /stress --restart=Never -n wsc2026 --overrides='{ "spec": { "tolerations": [{ "operator": "Exists" }], "node Selector": { "wsc2026/node": "application" }, "containers": [{ "name": "stress-mem", "image": "polinux/stress ", "resources": { "requests": { "memory": "64Mi" }, "limits": { "memory": "64Mi" } }, "command": ["stress", "--vm", " 1", "--vm-bytes", "60M", "--vm-keep", "-t", "3600"]] } } }' &>/dev/null sleep 180 GRAFANA_LB=\$(kubectl get svc -n observability -o jsonpath='{range .items[?(@.spec.type=="LoadBalance r")]}{.status.loadBalancer.ingress[0].hostname}{end}' 2>/dev/null) ... 다음 페이지 이어서 </pre>

순번	채점 항목
11-3	2) 출력된 Grafana URL을 통해 admin/Skills\$#\$@!에 로그인 후 wsc2026-grafana-dashboard에서 메트릭과 로그 형식이 채점자 사진과 일치하는지 확인하며, 모든 메트릭 및 로그는 빈값이 없어야 합니다. 다음 구성 요소가 대쉬보드에 포함되어 있는지 확인합니다. - Node CPU (%) - 노드별 CPU 사용률 - Node Memory (%) - 노드별 Memory 사용률 - Available Nodes - 각 노드그룹별 노드 수 - Pod CPU - 파드별 CPU 사용률 - Pod Memory - 파드별 Memory 사용률 - Pending Pods - Pending 상태의 Pod 개수 - Pod Restarts - 리스타트 Pod 개수 - App Pod CPU - 어플리케이션 파드별 CPU 사용률 - App Pod Memory - 어플리케이션 파드별 Memory 사용률 - App Running - 어플리케이션 파드 Running 상태의 수 - App Restarts - 어플리케이션 파드 리스타트 수 - App Pending - 어플리케이션 파드 Pending 상태의 수 - Request Count - 전체 어플리케이션 요청 수 - Response Time - 전체 어플리케이션 평균 응답 시간 - Status Code - 어플리케이션 응답 코드별 응답 수 - Application Logs - 어플리케이션 로그 - Application Logs 패널 로그 형식 예시 (/v1/book을 제외한 로그가 있을 경우 오답처리): INFO {"level":"INFO","path":"/v1/book","status":"200","duration":"112.663323ms","method":"POST"} (참고 사진) 

순번	채점 항목
11-4	1) Active Alerts 패널에 Edit을 선택해 나온 결과와 채점지 사진과 일치하는지 확인합니다. (* HighLatency Alert 구성요소는 채점과 무관합니다.) 